EDMS 30-315-1

SPECIFICATION

FOR

CARBON DIOXIDE

FIRE SUPPRESSION SYSTEM

Issue: August-2024 / Rev- 1

CONTENTS

1- SCOPE

This specification describes the minimum technical requirements for design, engineering, installation, testing, inspection and performance of CARBON DIOXIDE suppression system. The required system is intended to protect normally non occupied spaces and indoor network components of the Egyptian Electricity Distribution Companies (EDCs) against fire propagation.

2- APPLICABLE CODES and STANDARDS

Unless otherwise specified in this specification, the offered system should be designed, manufactured, and tested according to the latest editions of the relevant applicable codes and standards given in Table (1).

Table (1)

Standard No.	Description
NFPA12	Standard on CARBON DIOXIDE extinguishing system
NFPA 70	National Electrical Code.
NFPA 72E	National Fire Alarm.
ISO 14520	Gaseous – extinguishing systems-physical properties and system design – part 5-1-12 extinguishing.
ASME Publications American Society of Mechanical Engineering.	
IEEE Publication	
UL Publication Under Laboratories INC.	
The Egyptian Code for Design Basic and Implementation –Requirements to protect Facilities from Fire.	
CARBON DIOXIDE shall be replacement according to DOT (Department of Transportation) or similar requirements acceptable to the authority having jurisdiction.	

3- APROVAL AND LISTING

- Fire suppression system should be approved from the governmental authorities that certify compliance of the system with common engineering rules and the requirements of fire suppression systems.
- Typical approval / listing for fire suppression system:
  - UL LISTING or FM APPROVED or LPCB or VDS .

4- DEFINITIONS

- The required system includes a network of pipes distributed in places needed to be protected from the danger of fire. It is fed by compressed CARBON DIOXIDES gas cylinders which upon operation, rush through the sprinkler units under a certain pressure. Its particles are dispersed in a certain concentration in the protected space and wok to suffocate and extinguish the fire and prevent it.
- **Deep seated Fire:** A fire in ordinary combustible materials, such as wood, cloth, paper, rubber, and many plastics. A fire that involves energized electrical equipment Class A, C fire.
- **Surface fire:** A fire in flammable liquids, combustible liquids, petroleum greases, tars, oils, oil-based paints, solvents, lacquers, alcohols, and flammable gases class B fire.
- **Agent Concentration:**

  The portion of agent in an agent- air mixture expressed in volume percent.

5- LIMITATION OF USE

- All pre-engineered systems shall be installed to protect hazards within the limitations that have been established by the listing. Pre-engineered systems shall be listed to one of the following types:

- Those consisting of system components designed to be installed according to pre-tested limitations by a testing laboratory. These pre-engineered systems shall be permitted to incorporate special nozzles, flow rates, methods of application, nozzle placement, and pressurization levels that could differ from those detailed elsewhere in this standard. All other requirements of the standard shall apply.
- Automatic extinguishing units incorporating special nozzles, flow rates, methods of application, nozzle placement, actuation techniques, piping materials, discharge times, mounting techniques, and pressurization levels that could differ from those detailed elsewhere in this standard.

– CABON DIOXIDE shall not be used on fires involving the following materials unless the agents have been tested to the satisfaction of the authority having jurisdiction:
  - Certain chemicals or mixtures of chemicals, such as cellulose nitrate and gunpowder, which are capable of rapid oxidation in the absence of air.
  - Reactive metals such as lithium, sodium, potassium, magnesium, titanium, zirconium, uranium, and plutonium.
  - Metal hydrides
  - Chemicals capable of undergoing auto thermal decomposition, such as certain organic peroxides, pyrophoric materials, and hydrazine.

6- ENVIRONMENTAL CONDITIONS

- The performance of the offered clean agent should be guaranteed under the environmental and operational conditions given in Table (2). Any differences in the guaranteed performance should be clearly set out in the offer.

Table (2)

Minimum ambient temperature	-20°C
Maximum ambient temperature	100°C
Maximum relative humidity	95%
Maximum wind pressure	150 kg/mm
Storage temperature	10 to 60°C
System Lifetime	5 years
Battery Lifetime	5 years

7- TECHNICAL REQUIREMENTS

7.1 Electrical Operation Units:

- **Actuating Mechanism:**

  A mechanism whose automatic or manual operating leads to the discharge of extinguished agent.

- **Abort Switch:**

  A system control that, when operated during the releasing panels release delay count down, extends the delay in accordance with predetermined effect, provided that has UL Approval, and work includes the accessories necessary to install the switch and connect to the panel this and the work includes the accessories necessary to install the switch and connect to the panel.

- **Manual Release:**

  A means of manual release of the system shall be provided.

- **Control panel:**

  Control equipment shall be specifically for the number utilized, and their compatibility listed. Normal type Conventical consisted of 2 fire zones, provided that it has UL LISTED & FM APPROVED, and the work includes all fires retardant wires and EMT pipes necessary to connect the panel to all its component, to work efficiently.

– **Automatic detector:**

It is any listed method or device capable of detecting and indicating heat, smoke, or an abnormal condition in the hazard such as process trouble that is likely to produce fire. LPCB certificated. smoke and heat detectors provides that it is at least LPCB.

– **Alarm Bell:**

Installation 8-inch bell operating on voltages of 24 volts. It shall has LPCB. The work shall include all accessories necessary to install the switch and connect to the panel.

– **Congenital strobe / flasher buzzer:**

It shall be LPCB The work shall include all accessories necessary to install the switch and connect to the panel.

7-2-Agent Storage Containers.

– Carbon dioxides shall be stored in containers designed to hold that specific agent at ambient temperature.
– Containers shall be charged to a fill density or super pressurization level within the range specified in the manufacturer's listed manual.
– Agent container and gas must be UL LISTED or FM APPROVED or LPCB OR TUV.
– Cylinder shall be filled in the factory not Locally.
– Each CARBON DIOXIDES container shall have a permanent name- plate or other permanent marking that indicates the following:
    - For CARBON DIOXIDES containers, the agent, tare and gross weights, and

super pressurization level (where applicable) of the container.

- For CARBON DIOXIDES containers, the agent, pressurization level of the container, and nominal agent volume.

7-3 <u>Agent Supply</u>

– **<u>Primary Agent Supply:</u>**

– The quantity of CARBON DIOXIDES in the system primary agent supply shall be at least sufficient for the largest single hazard to be protected or group of hazards to be protected simultaneously.

– **<u>Reserve Agent Supply:</u>**

– Where required, a reserve agent supply shall consist of as many multiples of the primary agent supply as the authority having jurisdiction considers necessary?

7-4 <u>Storage Container Arrangement</u>

– Storage containers and accessories shall be located and arranged so that inspection, testing, recharging, and other maintenance activities are facilitated, and interruption of protection is held to a minimum.

– Storage containers shall be permitted to be located within or outside the hazard or hazards they protect.

– CARBON DIOXIDES storage containers shall not be located where they can be rendered inoperable or unreliable due to mechanical damage, exposure to chemicals or harsh weather conditions, or any other foreseeable cause. Where container exposure to such conditions is unavoidable, suitable enclosures or protective measures shall be employed.

8- <u>SYSTEM DESIGN</u>

8-1 Discharge Time

– <u>For surface fires</u>: the design concentration 34% shall be achieved within 1 minute from start of discharge.

- For deep-seated fires: the design concentration not less than 50% shall be achieved within 7 minutes, but the rate shall be not less than that required to develop a concentration of 30% at (2 minutes 30% at 2 minutes 40% at 3 minutes)

8-2 Openings That Cannot Be Closed

- Any openings that cannot be closed at the time of extinguishment shall be compensated for by the addition of a quantity of carbon dioxide equal to the anticipated loss at the design concentration during a 1-minute period.

  This amount of carbon dioxide shall be applied through the regular distribution system

8-3 Ventilating systems

- For ventilating systems that cannot be shut down, additional carbon dioxide shall be added to the space through the regular distribution system in an amount computed by dividing the volume moved during the liquid discharge period by the flooding factor.
- This additional amount of carbon dioxide shall be multiplied by the material conversion factor (given by a following Figure) when the design concentration is greater than 34% percent.

8-4 Agent Quantity

Total Flooding Quantity.

The quantity of CARBON DIOXIDE required to achieve the design concentration shall be calculated from the following equation:

W= V*flooding factor (f.f) *Correction factor

where: *W* = quantity of clean agent (kg)

V = net volume of hazard, calculated as the gross volume minus the volume of fixed structures impervious to clean agent vapor (m^3)

f.f =kg/m3(quantity of carbon dioxide

8-5 Minimum Carbon Dioxide Concentrations

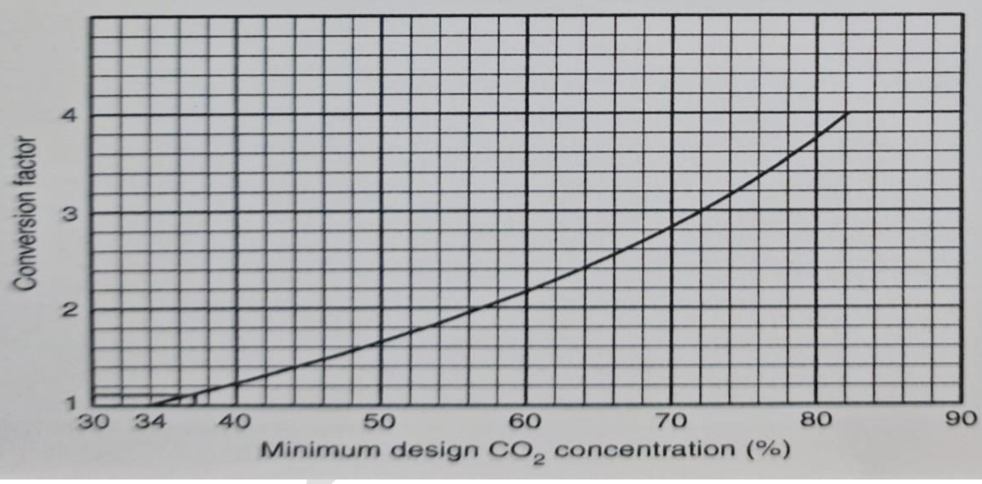
8-5-1 For surface fire:

The volume factor used to determine:

The basic quantity of carbon dioxides to protect an enclosure containing a material requiring a design concentration of 34% shall be in accordance with the following table.

Δ Table 5.3.3(b) Volume Factors and Flooding Factors (SI Units)

(A)	(B)		(C)
Volume of Space (m^3)	Volume Factor ($m^3/kg\ CO_2$)	Flooding Factor ($kg\ CO_2/m^3$)	Calculated Quantity (kg) (Not Less Than)
Up to 3.96	0.86	1.15	—
3.97–14.15	0.93	1.07	4.5
14.16–45.28	0.99	1.01	15.1
45.29–127.35	1.11	0.90	45.4
127.36–1415.0	1.25	0.80	113.5
Over 1415.0	1.37	0.74	1135.0



8-5-2 For Deep-Seated Fires.

Concentration shall be maintained for at least 20 minutes.

Table of volume factor and flooding for specific hazards:

Δ Table 5.4.2.1 Volume Factors and Flooding Factors for Specific Hazards

Design Concentration (%)	Volume Factors		Flooding Factors		Specific Hazards
	$ft^3/lb\ CO_2$	$m^3/kg\ CO_2$	$lb\ CO_2/ft^3$	$kg\ CO_2/m^3$	
50	10	0.62	0.100	1.60	Dry electrical hazards in general [spaces less than 2000 ft^3 (56.6 m^3)]
50	12	0.75	0.083 (200 lb minimum)	1.33 (91 kg minimum)	Dry electrical hazards in general [spaces greater than 2000 ft^3 (56.6 m^3)]
65	8	0.50	0.125	2.00	Record (bulk paper) storage, ducts, covered trenches
75	6	0.38	0.166	2.66	Fur storage vaults, dust collectors

9- ENCLOSURE

- In the design of a total flooding system, the characteristics of the protected enclosure shall be considered.
- The area of enclosable openings in the protected enclosure shall be kept to a minimum.
- The authority having jurisdiction shall be permitted to require pressurization depressurization of the protected enclosure or other tests to ensure performance that meets the requirements of this standard.
- To prevent loss of agent through openings to adjacent hazards or work areas, openings shall be permanently sealed or equipped with automatic closures by motorized damper or shut rover. Where reasonable confinement of agent is not practicable, protection shall be expanded to include the adjacent connected hazards or work areas, or additional agent shall be introduced into the protected enclosure using an extended discharge configuration Where CABON DIOXIDE total flooding system is being provided for the protection of a room with a raised or sunken floor, the room and raised or sunken floor shall be simultaneously protected.
- If only the space under the raised floor is to be protected by a total flooding system CARBON DIOXIDE shall be used to protect that space. Each volume, room, and raised or sunken floor to be protected shall be provided with detectors, piping network, and nozzles other than the ventilation system.
- Forced-air ventilating systems, including self-contained air recirculation systems, shall be shut down or closed automatically or by shut rover. Where their continued operation would adversely affect the performance of the fire extinguishing system or result in properly.

10- NOZZLES

- Discharge Nozzles shall be from the same manufacture of cylinders , Not Locally and shall be placed within the protected enclosure in compliance with listed limitations with regard to spacing, floor coverage, and alignment.
- The type, number and placement of nozzles shall be such that the design concentration will be established in all parts of the hazard enclosure and such that the discharge will not unduly splash flammable liquids or create dust clouds that could extend the fire, create an explosion, or otherwise adversely affect the contents or integrity of the enclosure.

10-1 Nozzle Selection

- The basis for nozzle selection shall be listed performance data that clearly depict the interrelation- ship of agent quantity, discharge rate, discharge time, area coverage, and the distance of the nozzle from the protected surface.
    - The maximum permitted time to extinguish a fire with a CARBON DIOXIDE shall be 60 seconds for surface fire and 7 minutes for deep seated fire (30% at 2 minutes 30% at 2 minutes 40% at 3 minutes).

10-2 Nozzle Discharge Rates

- The design discharge rate through individual nozzles shall be determined based on location or projection distance in accordance with specific approvals or listings.
- The system discharge rate shall be the sum of the individual rates of all the nozzles and discharge devices used in the system.

11- PIPE

- Pipe shall be of material having physical and chemical characteristics such that its integrity under stress can be predicted with reliability. Special corrosion-resistant materials or coatings shall be required in severely corrosive atmospheres. The thickness of the piping shall be calculated in accordance with ASME B31.1. The internal pressure used for this calculation shall not be less than the greater of the following values:
    - The normal charging pressure in the agent container at (21°C)
    - Eighty percent of the maximum pressure in the agent container at a maximum storage temperature of not less than (55°C), using the equipment manufacturer’s maximum allowable fill density, if applicable Pipe Connections
- Pipe joints other than threaded, welded, brazed, flared, compression, or flanged type shall be listed or approved.
- Fittings shall have a minimum rated working pressure equal to or greater than the minimum design working pressure specified, for the clean agent being used, or as other- wise listed or approved.
- For systems that employ the use of a pressure reducing device in the distribution piping, the fittings downstream of the device shall have a minimum rated working pressure equal to or greater than the maximum anticipated pressure in the downstream piping.
- Cast-iron fittings shall not be used.
- Class 150 fittings shall not be used.

12- TESTING, INSPECTION and MAINTENANCE

12-1 Preliminary Function Tests

- If the system is connected to an alarm receiving office, the alarm receiving office shall be notified that the fire system test is to be conducted and that an emergency response by the fire department or alarm station personnel is not desired.
- All personnel in areas that could be affected by the testing at the end user's facility shall be notified that a test is to be conducted.
- All personnel in areas that could be affected by the testing at the end user's facility shall be instructed as to events that could occur during testing of the fire extinguishing system.
- Each CARON Dioxide storage container release mechanism shall be disabled or replaced with a functional device so that activation of the release circuit will not release agent.

12-2 Routine Test:

  - Control panel shall be tested during manual release operation and detector.
  - All storage cylinder shall be tested during manual release operation.
- Each detector shall be tested for operation.
- All polarized alarm devices and auxiliary relays shall be checked for polarity in accordance with the manufacturer's instructions.
- Initiating and notification circuits shall be checked for end-of-line devices, if required.
- All supervised circuits shall be tested for trouble response.

12-3 Remote Monitoring Operation

- Each type of initiating device shall be operated while on standby power to verify that an alarm signal is received at the remote panel after the device is operated.

12-4 Functional operation test

- Each detection initiating circuit shall be operated to verify that all alarm functions occur according to design specifications.
- Each manual release shall be operated to verify that manual release functions occur according to design specifications.
- Each abort switch circuit shall be operated to verify that abort functions occur according to design specifications and that visual and audible supervisory signals are announced at the control panel.
- All automatic valves shall be tested to verify operation unless testing the valve will release agent or damage the valve (destructive testing).
- Pneumatic equipment, where installed, shall be tested for integrity to ensure operation.
- Integrity test must provide (if required).

13- NAMEPLATE

- Each CARBON DIOXIDE cylinder and accessories shall have a permanent nameplate that indicate the mass of agent – forming compound contained with manufacturer date, and date of mandatory replacement of generator based on useful life limit established in the listing. Certificate obtained such as:
    - At least LPCB Approved.
    - Any other certificate.
    - D.O.T (Department of Transport) or T.P.E.D (Transportable Pressure Equipment Directive).

14- GENERAL REQUIREMENTS

- The supplier acknowledges that he has fully inspected the work sites before bidding.

- The supplier shall provide the hydraulic calculations that confirm the requirements of this specification. The supplier shall submit all data tables furnished by the manufacturer to verify the adequacy of cylinders volume, nozzle type, pipes diameters and pipes distributions.
- The supplier shall submit all the catalogues and technical data of cylinders, nozzles and pipes.
- The supplier shall provide approval certificates of the system in accordance to the requirements of TPED, UL, FM and DOT.
- The supplier shall provide a reference list of similar projects.
- The supplier shall furnish CAD drawings describe the proposed network of fire suppression system and the distribution of all cylinders, nozzles and pipes.
- The contractor is responsible for supplying, installing and testing the system with all accessories necessary to complete the works. The contractor shall deliver the system in an acceptable manner in accordance to this contract and installation principles.
- The contractor shall return the working site to its original state before digging.
- All materials, equipment and devices shall meet the requirements of Egyptian civil defense, standards and the Egyptian Code.
- All network pipes shall be designated for fire networks (seamless, Table 40).
- All thermal cables shall be approved for fireworks. It shall be made of copper of 2×2 mm^2 covered with Shield filaments to protect the wire from tension.
- All fire alarm pipes shall be EMT of 20 mm diameter.
- All items shall be implemented in accordance with this technical specification, approved shop drawings, industry principles and instructions of _ _ EDC supervisors.

- The supplier must submit battery calculation.
- The supplier must submit filling ratio for EMT conduit.
- The supplier must submit CARBON DEOXIDES software shall be APPROVEL.
- The contractor guarantees the supplied system, materials and all accessories against any defects or malfunctions for a period of one year at least from the date of final delivery. The system should be monitored and maintained periodically during the warranty period.
- The system must operate in the presence of the agent of the brand provided.
- The contractor shall provide a certificate of origin of all part of the system.
- Approval certificate from Egyptian Electricity Holding Company should be submitted.

Alarm Bell

Abort switch

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 30-315-1
CARBON DIOXIDE Fire Suppression System	06-08-2024

Flasher buzzer

Manual release

Detector

Wires fire resistance

Control panel

نظام الغمر الكلي

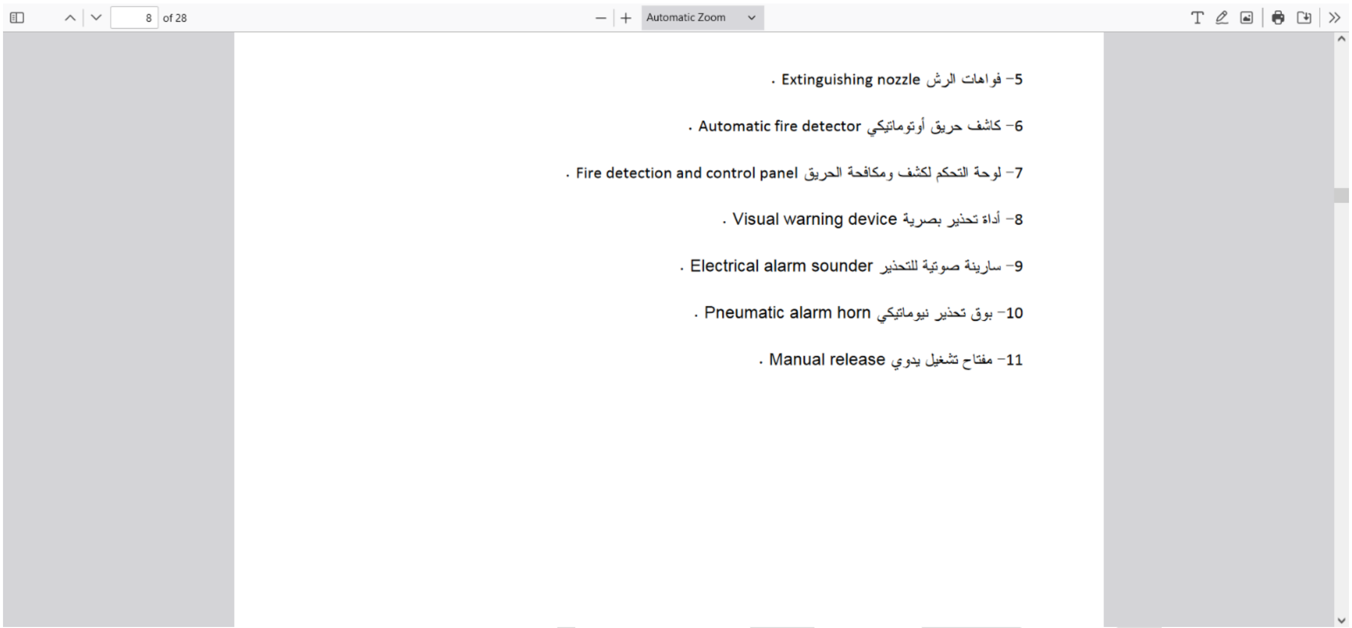
نظام الغمر الكلي

1- الأسطوانة المزودة بمشغل كهربي electrical actuator (Solenoid) .

2- أسطوانات مزودة بمشغل نيوماتيكي pneumatic actuation .

3 – تفريعة Manifold (خط التفريغ الرئيسي Main Discharge Line) .

4- أداة تأخير Delay device .


8 of 28 Automatic Zoom

5- فواهات الرش Extinguishing nozzle .

6- كاشف حريق أوتوماتيكي Automatic fire detector .

7- لوحة التحكم لكشف ومكافحة الحريق Fire detection and control panel .

8- أداة تحذير بصرية Visual warning device .

9- سارينة صوتية للتحذير Electrical alarm sounder .

10- بوق تحذير نيوماتيكي Pneumatic alarm horn .

11- مفتاح تشغيل يدوي Manual release .